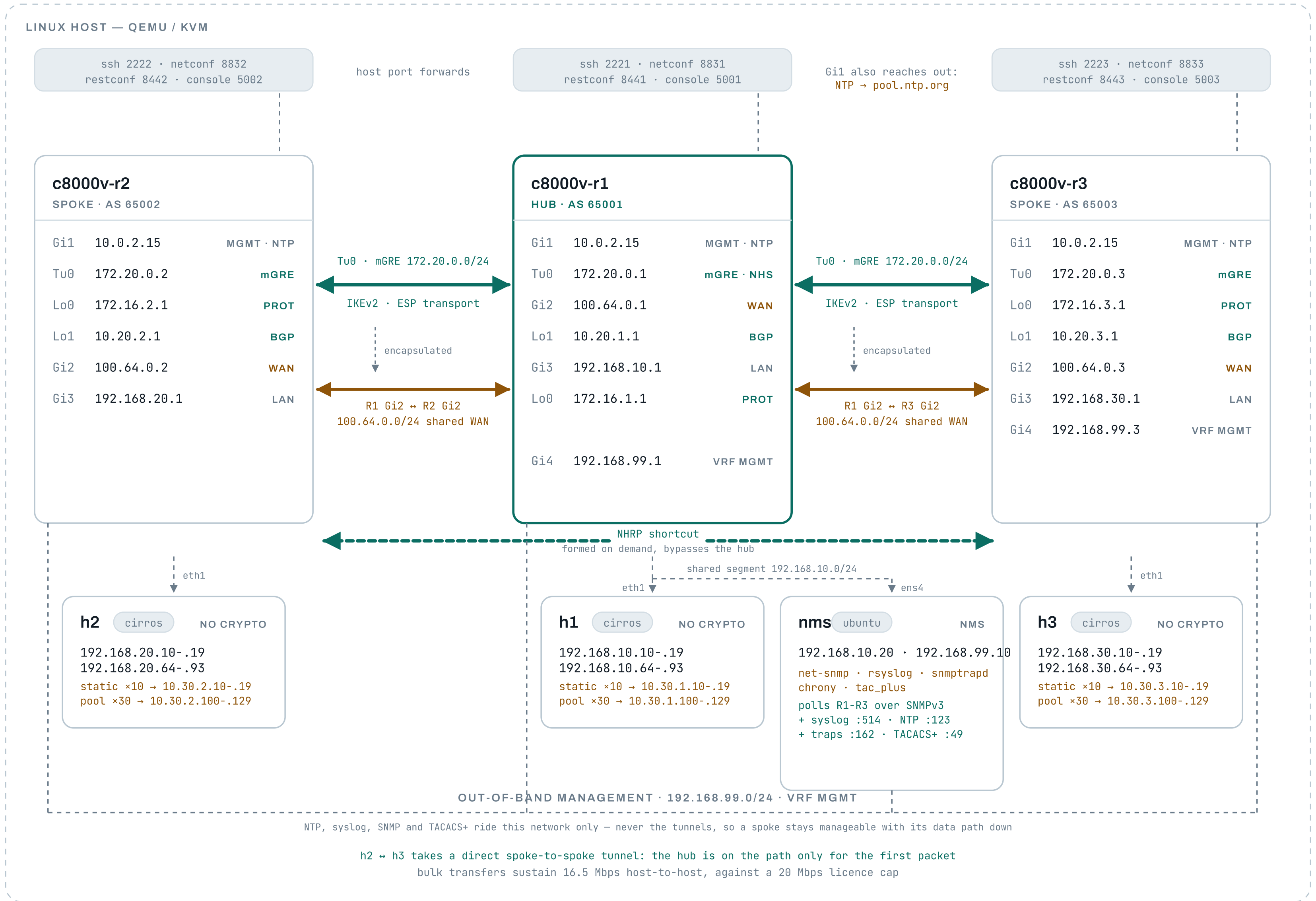
LAB REFERENCE

C8000V DMVPN Testbed

A DMVPN Phase 3 fabric of three Catalyst 8000V routers on one Linux host. One multipoint tunnel per router over a shared WAN, with NHRP supplying the destinations the configuration never names — so spokes build direct tunnels to each other on demand. Management runs entirely on a separate out-of-band network in its own VRF.

HOST	IMAGE	TOPOLOGY	COVERAGE	THROUGHPUT
Ubuntu 26.04 · QEMU 10.2 / KVM 3 routers + 3 hosts + NMS	c8000v-universalk9 _16G_serial.17.15.06	DMVPN Phase 3 AS 65001 / 65002 / 65003	208 tests · 22 suites Robot Framework	16.5 Mbps measured 20 Mbps licence cap



encrypted — ESP

cleartext — on the wire

encapsulation & management

Everything above each dashed arrow travels inside ESP; everything below it is readable on the link. The two connectors in each channel are the same physical path — a tunnel is an overlay that leaves the router through its link interface. The Linux hosts sit outside that boundary entirely and are unaware of it.

The hub carries a tunnel per spoke

R1 terminates `Tunnel0` to R2 and `Tunnel1` to R3 on separate physical interfaces, holding one IKEv2 SA and one BGP session per spoke. Each spoke holds exactly one of each.

Spokes never talk directly

Neither spoke has any route to the other's transport subnet — a test asserts this. h2 to h3 traffic is decrypted at the hub and re-encrypted onto the second tunnel, so it is protected on both legs.

Reachability comes from BGP, not statics

Each router advertises its own LAN and `Loopback1`. The hub re-advertises between spokes, so R2 learns R3's prefix with AS path `65001 65003` and a next hop of the hub's tunnel address.

Two kinds of source NAT, at scale

Every host carries 40 addresses. Ten (`.10-.19`) have their own static one-to-one mappings — deterministic, and reachable inbound. Thirty (`.64-.93`) are translated from a per-site pool of exactly thirty, `.100-.129`, allocated on demand. The pool has no `overload`, so each client takes its own address rather than sharing one by port: a correct run consumes the pool to 100% with no misses, and any sharing or failure is visible immediately.

Why translation is scoped by a route-map

Only traffic aimed at `10.30.0.0/16` is rewritten. Without that scoping both ends would rewrite their source on replies, and a ping would return from an address the sender never contacted. The direct LAN path therefore stays untranslated, and a test holds that line.

The overlay reorganises itself

Nothing configures a spoke with another spoke's address. The first packet between two of them goes through the hub, which answers with an NHRP redirect; the spokes resolve each other, install a shortcut, and the traffic moves to a direct tunnel protected by the same IPsec profile. The tests do not take that on trust: they clear the NHRP cache, prove the direct entry is absent, send traffic, and then require a traceroute between spokes to arrive in a single hop.

Throughput is licence-capped

The platform reports `20000 kb/s`. Bulk transfers between the hosts sustain 16.5 Mbps in both directions — and spoke-to-spoke is the slower path, since the hub decrypts and re-encrypts every packet. Raising the bar means raising the licensed level, not tuning the fabric.

nms collects logs and traps

All three routers send syslog and SNMPv3 traps to `nms`: the hub across its LAN, the spokes through the tunnels on routes BGP already carries. Each collector is a system service, so it is always receiving and no test run can leave a stale listener behind. These used to live on h1, which could only manage busybox `nc` — that attaches to its first sender and silently drops every other source, so each router needed a UDP port of its own for each service. rsyslog and snmptrapd each serve all three on one port.

Traps are received twice, deliberately

snmptrapd authenticates and decrypts each trap using a USM key created per router engine ID — proof the credentials, engine IDs and privacy protocol all agree, and the only way to see the varbinds. But it cannot show that a trap was encrypted *in flight*, because by the time it writes a log line it has already decrypted. So the routers also send to a raw capture on a second port, whose datagrams are BER-decoded byte by byte to assert the scoped PDU really is an opaque OCTET STRING and not a readable SEQUENCE.

Every BGP session is authenticated, tracked and filtered

MD5 on every neighbour; BFD on the tunnel each peer address lives on, with `fall-over bfd` so BGP acts on it; an outbound route-map per peer that both whitelists and tags — one clause per prefix class, setting a community of `<origin ASN>:<class>`, with the implicit deny as the filter; an inbound route-map per peer accepting only the nine communities this topology can produce, on an AS path that peer could legitimately send; and `maximum-prefix 15 80`, so a peer that floods takes itself out instead of being absorbed. A spoke may advertise only its own three prefixes (BGP /32, NAT domain, LAN); the hub may advertise its own plus the far spoke's, which is exactly what spoke-to-spoke

Encryption costs 62 bytes of every packet

ESP-AES-256 with SHA256 leaves 1438 bytes of the 1500-byte path. That boundary is where real IPsec deployments fail, and they fail quietly — small packets work, so ping and SSH look healthy while anything bulk stalls. The tests assert it from both sides: a packet of exactly 1438 crosses with DF set, 1439 does not, and both sizes succeed once fragmentation is allowed, which is what separates an MTU limit from a blackhole. Best of all, an end host two hops away learns `mtu=1438` for itself — path MTU discovery survives the tunnel, so a real client backs off instead of retransmitting into a hole.

Who may log in, and what they are told first

An `access-class` on the VTY lines limits SSH to the management network, enforced before authentication so an attempt from anywhere else never reaches a password prompt. It carries one documented exception — the hypervisor's user-mode NAT, which is how the test harness reaches these devices at all — pinned by a test asserting exactly two permits, so widening the policy fails rather than passes quietly. `vrf-also` is load-bearing and its absence is silent: without it the ACL is never consulted for sessions arriving through a VRF, and every out-of-band login is refused while the configuration reads perfectly correct. Ahead of all that sits a pre-login banner, captured by the tests from a needs. BFD timers are 500ms × 3 rather than sub-second: these routers share a host CPU, and an over-aggressive detector produces false positives. Blocking only BFD control packets, leaving BGP's TCP untouched, drops the session in about two seconds against a thirty-second hold time.

deliberately failed login to prove it reaches someone who has not been let in, and checked to disclose no vendor, model, version or hostname.

nms polls with real tooling

The cirros hosts cannot speak SNMP at all, which is why the trap tests decode BER by hand. `nms` is an Ubuntu cloud image with net-snmp, so polling is the same `snmpget` and `snmpwalk` an engineer would type. Its answers are cross-checked against each router's own CLI — sysName against the configured hostname, the interface walk against `show ip interface brief` — so a plausible reply from the wrong device cannot pass. Polling the spokes advances the ESP counters at both ends, which is how the tests prove the management traffic is itself encrypted.

Management is out of band, and in a VRF

Every router and the NMS share a flat `192.168.99.0/24`, on an interface held in the `MGMT` VRF — so the separation is enforced by the routing table rather than only by topology. The VRF knows how to reach the management network and nothing else; the global table has no route to it at all. The payoff is testable rather than theoretical: polling both spokes adds no measurable ESP traffic, because it never crosses a tunnel, and shutting a spoke's tunnel — which costs the hub its route to that spoke's LAN entirely — leaves the spoke still answering SNMP and still delivering syslog.

Time arrives as a hierarchy

The out-of-band network has no route to the internet, which is the point of it, so the routers cannot reach `pool.ntp.org` across it. Instead the NMS synchronises upstream through its own NAT interface and serves the management network with chrony: the public pool is still where time comes from, one stratum further away. That makes the server's own health load-bearing — chrony is configured `local stratum 10`, so it keeps serving whether or not it has an upstream, and without a test for that the whole lab could agree precisely on time that came from nowhere. Worth knowing too: IOS withholds its *synchronised* flag until the loopfilter finishes measuring drift, about 17 minutes, long after the offset is within milliseconds.

The underlay had to change

The static-VTI lab this mirrors gave each spoke a private /30 to the hub and asserted the spokes had *no* path to each other. DMVPN inverts that: a spoke can only build a direct tunnel if it can reach the far spoke's underlay address. So the two point-to-point links became one shared segment — which is what DMVPN models anyway, every site attached to a common provider cloud. A side effect worth having: with no listening sockets left, the routers can be started in any order.

Transport mode buys back 34 bytes

GRE already supplies an outer header, so IPsec runs in transport mode rather than adding a second one. The tunnel MTU is 1472 against the VTI lab's 1438: 28 bytes of overhead — 20 of outer IP, 4 of GRE, 4 for the tunnel key — instead of 62. Worth knowing too: the first packet across a new spoke-to-spoke path is consumed establishing it, so a cold measurement reports 80 percent and says nothing about the MTU.

Host access

SERVICE	c8000v-r1 (hub)	c8000v-r2	c8000v-r3
SSH	2221	2222	2223
NETCONF	8831	8832	8833
RESTCONF	8441	8442	8443
Serial console	telnet 5001	telnet 5002	telnet 5003
Host behind it	h1 · ssh 2231	h2 · ssh 2232	h3 · ssh 2233
Host console	telnet 5011	telnet 5012	telnet 5013
NMS	nms · ssh 2241 · telnet 5021	polled across the tunnels from the hub LAN	

Brought up from cold with `./start-lab.sh` — day0 bootstrap, licence reload, IPsec, eBGP, then the hosts. Validated by `./run-tests.sh`: 208 tests across 22 suites, including sustained throughput, 10 static NAT mappings and a 30-address dynamic pool per site, MD5-authenticated and BFD-tracked eBGP with community tagging and per-peer inbound validation, NTP against the public pool, syslog and SNMPv3 traps delivered to collectors on nms, which also polls every router over SNMPv3. Every run is archived under `results/results_<date>_<time>/`.